

Biswajit Mohapatra

AI ENGINEER | MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Dynamic and results-driven AI Engineer with 5+ years of experience leading the design, implementation, and deployment of production-grade, LLM-powered AI solutions for large enterprise clients in regulated environments. Proven expertise in translating complex business challenges (Financial Services, Health) into scalable AI architectures, specifically leveraging **LLM/RAG systems**, Agentic workflows, and MLOps principles. Adept at **client-facing delivery**, solution design, and establishing secure, governed systems using technologies like **AWS, Docker, and CI/CD**. Seeking to drive data- and technology-led transformation by delivering robust, scalable AI solutions at the enterprise level.

AREAS OF EXPERTIES

AI & LLM Engineering : LLM/RAG Implementation | Prompt Engineering | Retrieval-Augmented Generation (RAG) | Agentic AI Frameworks (LangChain) | Agent Harness | LLM Finetuning | LLM Distillation

MLOps & Deployment: CI/CD Pipelines | MLFlow | Model Versioning | AI Observability & Monitoring

Cloud & Architecture: AWS | Azure | DataBricks | CloudFormation | Terraform

Engineering & Tools: Python | Docker | Kubernetes | AWS / Azure | APIs | Microservices | Git | FastAPI | Vector Databases | SQL

PROFESSIONAL EXPERIENCE

Senior GenAI Engineer | Takeda Pharmaceuticals | Bengaluru, Karnataka October 2025 – Present

- Led Cross-Functional AI Platform Delivery: Architected and deployed an end-to-end AI-powered analytics platform utilizing LLM inferencing (AWS Bedrock) and conversational BI to process unstructured data. This solution drove actionable insights into donor behavior, directly contributing to a 13% increase in donor retention and aligning with key business objectives.
- Established Production-Grade Safety Guardrails: Directed the fine-tuning and release of a Llama 3.2 3B model specifically for PII/PHI detection, establishing a crucial safety layer for 20+ enterprise AI solutions. This initiative established a repeatable, secure methodology for deploying sensitive models in a regulated environment.
- Agentic SDLC Framework: Led a team of three engineers in designing and implementing an enterprise-ready Agentic Software Development Lifecycle (SDLC) framework. Successfully integrated concepts like Agent Harness, context engineering, and LLM hand-off to create a scalable, standardized methodology for Agentic workflows across the organization.
- Ensured MLOps and Automation: Established robust MLOps pipelines by integrating Databricks for development, MLFlow for comprehensive experiment tracking, and GitHub Actions for CI/CD. This ensured the AI platform was built with enterprise-grade automation, version control, and seamless deployment capabilities.
- Strategic Product Innovation & Stakeholder Influence: Conceptualized and drove the vision for an Agentic AI-powered Donor Engagement system. Analyzed multi-source data (appointment history, visit history, payment, feedback) to solve critical drop-off issues (33% signup failure, 40% scheduling failure). Presented this solution—a personalized slot recommendation system via a seamless chat interface—to the Head of Biolife, resulting in its inclusion in the 2026 MRP.
- Strategic Proof of Concept (PoC) Delivery: Initiated and delivered strategic PoCs focused on operational efficiency, including a Conversational BI system for workforce adherence and an AI-powered L1 support system for compliance, demonstrating the ability to translate ambiguous operational challenges into high-value, data-driven solutions for stakeholders.
- Team Enablement & Mentorship: Acted as a technical leader by facilitating internal GenAI community sessions, successfully educating teams on AI adoption. Mentored junior engineers and analysts on adopting Databricks for advanced data analysis and AI development.

Senior Analyst | Ernst & Young | Bengaluru, Karnataka July 2024 – September 2025

- Delivered Financial Risk Mitigation System: Led the end-to-end delivery of an AI-powered invoice management system, automating validation that prevented overpayments exceeding \$2M annually, demonstrating a direct impact on financial risk management for the client.
- Engineered LLM-Powered Knowledge Retrieval (RAG): Architected and implemented a RAG-based document retrieval system (SOP Navigator) utilizing AWS Bedrock for LLM inference and Databricks for vector storage. This solution eliminated manual PDF searches, streamlined compliance workflows, and improved operational efficiency across teams.
- Developed Conversational AI Workbench: Built and launched a conversational interface that allowed users to query complex work order data using natural language. This initiative cut manual effort by 10 hours per week while ensuring enhanced accuracy and timeliness in reporting.
- Created Autonomous Data Generation Agent: Engineered an Agentic AI-powered synthetic data generation tool that autonomously analysed ICD-10-CM codes to cross-check and validate information from knowledge bases, ensuring high data accuracy and reliability for critical medical data.

Associate Data Science Engineer | Iamneo Edutech | Coimbatore, Tamil Nadu
June 2022 – June 2024

Key contributions:

- Led a team to design, develop, and launch Neo-screener, an AI-based interview and video assessment product, resulting in a 38% sales increase for Iamneo after launch. Used technologies like FastAPI for API development, SQL for data storage, PyTorch for custom training of language models, MLFlow for pipeline management, Docker for containerization, Azure Container Registry for version control, and Azure Kubernetes for deployment. Directed a cross-functional team, ensuring successful project completion on time and within budget while fostering innovation and teamwork.
- Developed automated pipelines for analytical report generation, reducing manual work by 20 hours per week and improving data accuracy and reporting timeliness. Utilized GitHub Workflows for automation, Google Cloud Functions for serverless execution, and Google BigQuery and MySQL for data management. The automation improved report reliability and frequency, boosting team productivity and facilitating better data-driven decision-making.

Data Science Intern | Ineuron.ai | Bangaluru, Karnataka
March 2021 – May 2022

As a Data Science Intern at Ineuron, I played a pivotal role in sustaining automation pipeline code bases, analyzing business datasets, and delivering ad-hoc analytical reports.

Key contributions:

- Diligently maintained and addressed critical bugs in diverse production automation pipelines and data processing pipelines.
- Conducted in-depth analysis of business datasets, uncovering valuable insights, and effectively communicated findings to the stakeholders.

EDUCATION

Bachelors of Technology | GIET
University CGPA: 8.24 YOP: 2021

Higher Secondary Education | Gandhi Public
School CGPA: 6.60 YOP: 2017

Secondary Education | Gandhi Public
School CGPA: 8.40 YOP: 2015

CERIFICATIONS

Databricks Generative AI Engineer Associate
Provider: Databricks

Building RAG Agents with LLM

Provider: Nvidia

Fundamentals of Generative AI Databricks

Provider: Databricks

Full Stack Data Science Certification
Provider: Ineuron.ai

Artificial Intelligence Operations
Provider: Ineurion.ai

